Lagrange Interpolation & The Chinese Remainder Theorem

Tsz-Wo Sze

August 9, 2025

Abstract

In this article, we present an application of Lagrange interpolation to solve the puzzle below. Then, we discuss the similarity between Lagrange interpolation and the Chinese remainder theorem.

$$\begin{aligned} 37\#21 &= 928, \\ 77\#44 &= 3993, \\ 123\#17 &= 14840, \\ 71\#6 &= \quad ? \end{aligned}$$

1 Lagrange Interpolation

Theorem 1.1 (Lagrange Interpolation)**.** *Let $(a_i, b_i) \neq (0,0)$ be n distinct pairs of numbers and c_i be n not necessarily distinct numbers. By Lemma 1.2 below, we assume*

$$a_i b_j - a_j b_i \neq 0 \qquad \text{for all } i \neq j. \tag{1.1}$$

Define the Lagrange polynomial

$$f(x,y) := \sum_{1\leq i\leq n} c_i \prod_{1\leq j\leq n, j\neq i} \frac{x b_j - a_j y}{a_i b_j - a_j b_i}. \tag{1.2}$$

Then, f is a degree $d < n$ homogeneous polynomial with $f(a_i, b_i) = c_i$ for all $1 \leq i \leq n$.

Lemma 1.2. *Suppose $a_{i_0} b_{j_0} - a_{j_0} b_{i_0} = 0$ for some $i_0 \neq j_0$. By translatiing a_i's and b_i's, let*

$$A_{i,s} := a_i + s \qquad \text{and} \qquad B_{i,t} := b_i + t.$$

There exists integers $1 \leq s \leq n$ and $1 \leq t \leq n(n-1)/2$ such that $A_{i,s} B_{j,t} - A_{j,s} B_{i,t} \neq 0$ for all $i \neq j$.

Proof. If $a_i \neq 0$ for all i, set $s = 0$. Otherwise, there are at most $n-1$ non-zero a_i's. Choose $1 \leq s \leq n$ such that $s \neq -a_j$ for $1 \leq j \leq n$. Then, $A_j \neq 0$ for all j. We may assume $a_i \neq 0$ for all i.

If $a_{i_0} = a_{j_0}$, then $b_{i_0} = b_{j_0}$ and these two pairs are not distinct. We may assume $a_{i_0} \neq a_{j_0}$. Then,

$$a_{i_0} B_{j_0,t} - a_{j_0} B_{i_0,t} = (a_{i_0} - a_{j_0}) t \neq 0 \qquad \text{for } 1 \leq t \leq n(n-1)/2.$$

If $a_i B_{j,t_0} - a_j B_{i,t_0} = 0$ for some $(i,j) \neq (i_0, j_0)$, then $a_i B_{j,t_1} - a_j B_{i,t_1} = (a_i - a_j)(t_1 - t_0) \neq 0$ for any $t_1 \neq t_0$. Since there are exactly $n(n-1)/2 - 1$ pairs of i, j other than i_0, j_0, the lemma follows. □

1.1 Solving The Puzzle

The puzzle mentioned in the abstract has $n = 3$. Using the format $a_i \# b_i = c_i$, let $a_1 = 37, b_1 = 21, c_1 = 928, a_2 = 77, b_2 = 44, c_2 = 3993, a_3 = 123, b_3 = 17, c_3 = 14840$. Then,

$$\begin{aligned} a_1 b_2 - a_2 b_1 &= 37 \cdot 44 - 77 \cdot 21 &&= 11, \\ a_1 b_3 - a_3 b_1 &= 37 \cdot 17 - 123 \cdot 21 &&= -1954, \\ a_2 b_3 - a_3 b_2 &= 77 \cdot 17 - 123 \cdot 44 &&= -4103. \end{aligned}$$

Let $f(x,y) := x\#y$. Then,

$$\begin{aligned} f(x,y) &= \frac{c_1(xb_2 - a_2 y)(xb_3 - a_3 y)}{(a_1 b_2 - a_2 b_1)(a_1 b_3 - a_3 b_1)} + \frac{c_2(xb_1 - a_1 y)(xb_3 - a_3 y)}{(a_2 b_1 - a_1 b_2)(a_2 b_3 - a_3 b_2)} + \frac{c_3(xb_1 - a_1 y)(xb_2 - a_2 y)}{(a_3 b_1 - a_1 b_3)(a_3 b_2 - a_2 b_3)} \\ &= 928 \frac{(44x - 77y)(17x - 123y)}{(11)(-1954)} + 3993 \frac{(21x - 37y)(17x - 123y)}{(-11)(-4103)} + 14840 \frac{(21x - 37y)(44x - 77y)}{(1954)(4103)} \\ &= x^2 - y^2. \end{aligned}$$

Finally, $71\#6 = 71^2 - 6^2 = 5005$.

1.2 The Idea Behind Lagrange Interpolation

The Lagrange polynomial (1.2) looks complicated but the idea behind it actually is simple. Given n distinct pairs $(a_i, b_i) \neq (0,0)$ and n numbers c_i, construct a homogeneous polynomial f satisfying $f(a_i, b_i) = c_i$. For discussion purpose, set $n = 3$. Define

$$f(x,y) := c_1\delta_1(x,y) + c_2\delta_2(x,y) + c_3\delta_3(x,y),$$

where δ_i is a homogeneous polynomial having the following property:

$$\delta_i(x,y) := \begin{cases} 1 & \text{, if } x = a_i \text{ and } y = b_i, \\ 0 & \text{, if } x = a_j \text{ and } y = b_j \text{ for } j \neq i. \end{cases} \tag{1.3}$$

For $i = 1, 2, 3$, we have $f(a_i, b_i) = c_i$ as desired. The remaining task is to construct δ_i.

Take δ_1 as an example. Since we want to have $\delta_1(a_2, b_2) = 0$ and $\delta_1(a_3, b_3) = 0$, define it as

$$\delta_1(x,y) := (xb_2 - a_2y)(xb_3 - a_3y)k_1,$$

for some constant k_1. Then,

$$\delta_1(x,y) = \begin{cases} (a_1b_2 - a_2b_1)(a_1b_3 - a_3b_1)k_1 & \text{, if } x = a_1 \text{ and } y = b_1, \\ 0 & \text{, if } x = a_j \text{ and } y = b_j \text{ for } j \neq 1. \end{cases}$$

By assumption (1.1), we have $(a_1b_2 - a_2b_1)(a_1b_3 - a_3b_1) \neq 0$. Set $k_1 := (a_1b_2 - a_2b_1)^{-1}(a_1b_3 - a_3b_1)^{-1}$ in order to have $\delta_1(a_1, b_1) = 1$. Therefore,

$$\delta_1(x,y) = \frac{(xb_2 - a_2y)(xb_3 - a_3y)}{(a_1b_2 - a_2b_1)(a_1b_3 - a_3b_1)}$$

satisfies property (1.3). Of course, δ_2 and δ_3 can be constructed similarly.

2 Chinese Remainder Theorem

Theorem 2.1 (Chinese Remainder Theorem)**.** *Let $a_1, \ldots, a_n > 1$ be n pairwise coprime integers and $A = \prod_{1 \leq i \leq n} a_i$. Let m be an integer such that*

$$m \equiv c_1 \pmod{a_1}, \qquad \cdots, \qquad m \equiv c_n \pmod{a_n}. \tag{2.1}$$

Then,

$$m \equiv \sum_{1 \leq i \leq n} c_i \prod_{1 \leq j \leq n, j \neq i} a_j k_{i,j} \pmod{A}, \qquad \text{where } k_{i,j} \equiv a_j^{-1} \pmod{a_i}. \tag{2.2}$$

2.1 The Idea Behind The Chinese Remainder Theorem

Interestingly, the Chinese remainder theorem equation (2.2) looks similar to the Lagrange polynomial (1.2). Indeed, the ideas behind them are essentially the same. As before, we set $n = 3$ in the following discussion. In order to satisfy conditions (2.1), define

$$m := c_1\delta_1 + c_2\delta_2 + c_3\delta_3,$$

where δ_i is an integer such that

$$\delta_i \bmod a_j \equiv \begin{cases} 1 & \text{, if } j = i, \\ 0 & \text{, if } j \neq i. \end{cases} \tag{2.3}$$

For $i = 1, 2, 3$, we have $m \equiv c_i \pmod{a_i}$ as desired. The remaining task is to construct δ_i.

Take δ_1 as an example. Since we want to have $\delta_1 \equiv 0 \pmod{a_2}$ and $\delta_1 \equiv 0 \pmod{a_3}$, define it as

$$\delta_1 := a_2a_3k_1,$$

for some integer k_1. Then,

$$\delta_1 \equiv \begin{cases} a_2a_3k_1 & \text{, if } j = i, \\ 0 & \text{, if } j \neq 1. \end{cases}$$

By the assumption of a_i's being pairwise coprime, let $k_{1,2} \equiv a_2^{-1} \pmod{a_1}$ and $k_{1,3} \equiv a_3^{-1} \pmod{a_1}$. Set $k_1 := k_{1,2}k_{1,3}$ in order to have $\delta_1 \equiv 1 \pmod{a_1}$. Therefore,

$$\delta_1 = a_2a_3k_{1,2}k_{1,3}$$

satisfies property (2.3). Of course, δ_2 and δ_3 can be constructed similarly.